

Distinct exponents give no logarithm for every (k, h) :

$$Z(n) \sim \frac{b^{k_2(p_2-p_1)} C(p_1, p_2)}{k_1 k_2} n^{-p_1/2}$$

Claude, with Daniel Murfet

2026-09-06

Abstract

The third regime of the general two-dimensional reduction (unit 44). For $p_1 < p_2$ the noncritical integral $C(p_1, p_2) = \int_0^\infty x^{p_1-p_2-1} F_{p_2-1}(x) dx$ converges to a positive constant, and

$$Z(n) \sim \frac{1}{k_1 k_2} b^{k_2(p_2-p_1)} C(p_1, p_2) n^{-p_1/2},$$

the multiplicity-one prediction of `thm:TaylorTree` with sharp exponent $p_1/2$. At $k = (1, 1)$, $h = (0, 1)$ this is unit 37 (there $C = A$). Zero `sorry/axiom`.

1 Setting

Convergence of C needs two dominations from unit 45: near 0, $F_{p_2-1}(x) \leq E x^{p_2}/p_2$ makes the integrand $\leq (E/p_2) x^{p_1-1}$, integrable because $p_1 > 0$; near ∞ , $F_{p_2-1} \leq A_{p_2-1}$ makes it $\leq A_{p_2-1} x^{p_1-p_2-1}$, integrable because $p_1 - p_2 - 1 < -1$. Positivity of C is positivity of the integrand on $(0, \infty)$, which in turn needs $F_\gamma(x) > 0$ for $x > 0$.

2 The things formalised

```

theorem weightedPrimitive_pos ... (hx : 0 < x) : 0 < weightedPrimitive a x
theorem noLog_integrand_integrableOn ( a p p : ) (h : 0 < ) (hp : 0 < p )
  (hlt : p < p ) :
  IntegrableOn (fun x => x ^ (p - p - 1) * weightedPrimitive a (p - 1) x) (Ioi 0)
noncomputable def noLogConst ( a p p : ) : :=
  x in Ioi 0, x ^ (p - p - 1) * weightedPrimitive a (p - 1) x
theorem noLogConst_pos ... : 0 < noLogConst a p p
theorem twoDGeneral_isEquivalent_of_lt ...
  (hlt : ((h : ) + 1) / k < ((h : ) + 1) / k ) :
  (fun n => twoDGeneral a b n h h k k ) ~[atTop]
  fun n => 1 / ((k : ) * k )
    * b ^ ((k : ) * ((h : ) + 1) / k - ((h : ) + 1) / k ))
    * noLogConst a (((h : ) + 1) / k ) (((h : ) + 1) / k )
    * n ^ (-(((h : ) + 1) / k / 2))

```

3 Proof strategy

Integrability on $(0, \infty)$ is assembled from $(0, 1]$ and $(1, \infty)$ with the two dominations and `IntegrableOn.union`. The truncated integral converges to C by Mathlib’s `intervalIntegral_tendsto_integral_Ioi` along $L_n = \sqrt{n} b^{k_1+k_2} \rightarrow \infty$. The asymptotic then follows without any explicit remainder bound: the truncated integral is `IsEquivalent` to the constant C (`isEquivalent_const_iff_tendsto`, valid because $C \neq 0$), and multiplying by the exact prefactor $\frac{1}{k_1 k_2} b^{k_2(p_2-p_1)} n^{-p_1/2}$ via `IsEquivalent.mul` gives the result after two pointwise `congr` steps. The exponents are again introduced as opaque variables to keep `ring` honest. Numerically the convergence is slow when $p_2 - p_1$ is small (the tail decays like $L^{-(p_2-p_1)}$): at $k = (2, 3)$, $h = (0, 1)$ the ratio reaches 1 only around $n \approx 10^9$, which the one-dimensional integral confirms exactly.

4 Roadblocks and resolutions

- None in Lean: the file compiled on the first pass. The only care needed was on the numerics, where a two-dimensional quadrature at moderate n suggested a discrepancy that was in fact the slow $L^{-1/6}$ tail; checking the one-dimensional truncated integral at $n = 10^9$ resolved it.

5 What was Mathlib, what was new

Mathlib: `intervalIntegral_tendsto_integral_Ioi`, `isEquivalent_const_iff_tendsto`, `IsEquivalent.mul`, `setIntegral_pos_iff_support_of_nonneg_ae`, `integrableOn_Ioi_rpow_of_lt`, `IntegrableOn.union`. New: the noncritical constant $C(p_1, p_2)$ and the general no-logarithm asymptotic.

6 Lessons

When the limit is an honest convergent integral rather than a closed form, the `IsEquivalent`-to-a-constant route is far shorter than an explicit tail bound. And a slow numerical drift toward the predicted constant is not evidence against the theorem: check the rate the analysis predicts before suspecting the formula.

7 Outcome

All three regimes of the general two-dimensional standard integral are now formalised: equal exponents (unit 45, one logarithm), $p_1 < p_2$ (this unit, no logarithm, exponent $p_1/2$) and, by symmetry of the roles of the coordinates, $p_1 > p_2$. Together with the exact reduction of unit 44 this closes the two-dimensional theory of `thm:TaylorTree`’s leading term for arbitrary (k, h) . Next candidates: the $p_1 > p_2$ case stated directly, the mixed-multiplicity case in general dimension, or the next-order term in the general equal case.